BUS 104: Practice Set — Integer Linear Programming (ILP)

Problems and Solutions (Practice Set 3: LP Sensitivity Analysis)

Problem 4-8: Detecting Alternate Optimal Solutions

How do we detect the presence of alternate optimal solutions from a Solver Sensitivity Report?

Solution to Problem 4-8

In most cases, when the **Allowable Increase** or **Allowable Decrease** column for the objective function coefficient (OFC) of a decision variable has a value of **zero** in the *Adjustable Cells* table of the Solver Sensitivity Report (assuming the solution is non-degenerate), this indicates the presence of **alternate optimal solutions**. Equivalently, if a non-basic variable (a variable with a `Final Value` of zero) has a **Reduced Cost** of zero, alternate optimal solutions exist.

Problem 4-15: Whole Food Nutrition Center

Consider the Whole Food Nutrition Center diet problem example discussed in section 3.7 (page 107) of Chapter 3. Use the Sensitivity Report for this LP model (shown in Screenshot 4-8) to answer the following questions. Each question is independent of the others.

Screenshot 4-8: Solver Sensitivity Report for Problem 4-15 (Whole Food Nutrition Center)

Adjustable Cells

Cell	Name	Final Value	Reduced Cost	Obj. Coeff.	Allow. Inc.	Allow. Dec.
B5	Number of pounds Grain A	0.025	0.000	0.330	0.063	1E+30
C5	Number of pounds Grain B	0.050	0.000	0.470	1E+30	0.190
D5	Number of pounds Grain C	0.050	0.000	0.380	1E+30	0.073

Constraints

Cell	Name	Final Value	Shadow Price	RHS	Allow. Inc.	Allow. Dec.
E8	Protein	3.000	0.038	3.000	0.000	0.250
E9	Riboflavin	2.350	0.000	2.000	0.350	1E+30
E10	Phosphorus	1.000	0.088	1.000	0.016	0.000
E11	Magnesium	0.425	0.000	0.425	0.000	1E+30
E12	Total Mix	0.125	-1.210	0.125	0.004	0.000

(a) What is the impact if the daily allowance for protein can be reduced to 2.9 units?

(b) Whole Food believes the unit price of grain A could be 5% overestimated and the unit price of grain B could be 10% underestimated. If these turn out to be true, what are the new optimal solution and optimal total cost?

(c) What is the impact if the reduction in the daily allowance for protein in (a) requires Whole Food to simultaneously increase the daily allowance of riboflavin to 2.20 units?

Solution to Problem 4-15

Note: At the current optimal solution ($G_A = 0.025$ lbs, $G_B = 0.050$ lbs, $G_C = 0.050$ lbs), the baseline total cost is $\$0.33(0.025) + \$0.47(0.050) + \$0.38(0.050) = \$0.05075$.

(a) Reducing Protein Requirement to 2.9 Units: Reducing the daily allowance of protein from 3.000 to 2.900 units is a decrease of $\Delta\text{RHS} = 0.10$ units, which is within the allowable decrease of 0.250 units. Using the shadow price of \$0.038, the total cost will **decrease by** $0.10 \times \$0.038 = \$\mathbf{0.0038}$ (to $\$0.05075 - \$0.0038 = \$0.04695$).

(b) Simultaneous Changes in Grain A and Grain B Prices:

- **Revised cost of Grain A:** \$0.33/1.05 = \$0.314/lb, a decrease of $\Delta c_A = \$0.016$ (allowable decrease $= \infty$).
- **Revised cost of Grain B:** \$0.47/0.90 = \$0.522/lb, an increase of $\Delta c_B = \$0.052$ (allowable increase $= \infty$).

Checking the **100% rule** for simultaneous OFC changes: $(0.016/\infty) + (0.052/\infty) = 0 < 1$. Therefore, the **current optimal solution remains optimal** (0.025 lbs Grain A, 0.050 lbs Grain B, 0.050 lbs Grain C), and the **new total cost** is $\$0.314(0.025) + \$0.522(0.050) + \$0.380(0.050) = \$0.05295 \approx \$\mathbf{0.0529}$ (or \$0.0527 if computed as $0.95 \times \$0.33 = \$0.3135$ and $1.10 \times \$0.47 = \$0.517$).

(c) Simultaneous Changes in Protein and Riboflavin Allowances: Protein decreases by 0.10 units (allowable decrease = 0.250) and riboflavin increases by 0.20 units (allowable increase = 0.350). Checking the **100% rule**: $(0.10/0.25)+(0.20/0.35) = 0.40+0.5714 = 0.9714 < 1$. Thus, the shadow prices remain valid, and total cost changes by $(0.20 \times \$0) - (0.10 \times \$0.038) = -\$\mathbf{0.0038}$ (a **decrease of \$0.0038**).

Problem 4-16: Quitmeyer Electronics Incorporated

Consider the Quitmeyer Electronics product mix problem first presented in Chapter 3 as Problem 3-3 on page 125. Use Solver to create the Sensitivity Report for this LP problem. Then answer the following questions using this report. Each question is independent of the others.

(a) Interpret the reduced costs for the products that are not currently included in the optimal production plan.
(b) Another part of the corporation wants to take 35 hours of time on test device 3. How does this affect Quitmeyer's optimal solution?
(c) Quitmeyer has the opportunity to obtain 20 additional hours on test device 1 at a cost of \$25 per hour. Is this deal worthwhile?
(d) Quitmeyer has the opportunity to give up 20 hours of time on device 1 and obtain 40 hours of time on device 2 in return. Is this deal worthwhile? Justify your answer.

Reference Data from Problem 3-3 (Page 125): Quitmeyer manufactures six devices: internal modems (I), external modems (E), circuit boards (C), floppy drives (F), hard drives (H), and memory boards (M). Test devices 1 and 2 have 120 hrs/week; device 3 has 100 hrs/week. Variable labor costs are \$15/hr on device 1, \$12/hr on device 2, and \$18/hr on device 3.

Item	**Internal (I)**	**External (E)**	**Circuit (C)**	**Floppy (F)**	**Hard (H)**	**Memory (M)**
Rev. / Mat. cost	\$200 / \$35	\$120 / \$25	\$180 / \$40	\$130 / \$45	\$430 / \$170	\$260 / \$60
Dev. 1, 2, 3 (min)	7, 2, 5	3, 5, 1	12, 3, 3	6, 2, 2	18, 15, 9	17, 17, 2
Labor cost/unit	\$3.65	\$2.05	\$4.50	\$2.50	\$10.20	\$8.25
Net Profit/unit	**\$161.35**	**\$92.95**	**\$135.50**	**\$82.50**	**\$249.80**	**\$191.75**

Solution to Problem 4-16

1. LP Formulation and Excel Sensitivity Report (see `P4-16.xls`)

$$\text{Maximize Profit} = 161.35I + 92.95E + 135.50C + 82.50F + 249.80H + 191.75M$$

subject to testing time constraints (in hours, dividing minutes per unit by 60):

$$\tfrac{7}{60}I + \tfrac{3}{60}E + \tfrac{12}{60}C + \tfrac{6}{60}F + \tfrac{18}{60}H + \tfrac{17}{60}M \le 120 \quad \text{(Test device 1 hours)}$$
$$\tfrac{2}{60}I + \tfrac{5}{60}E + \tfrac{3}{60}C + \tfrac{2}{60}F + \tfrac{15}{60}H + \tfrac{17}{60}M \le 120 \quad \text{(Test device 2 hours)}$$
$$\tfrac{5}{60}I + \tfrac{1}{60}E + \tfrac{3}{60}C + \tfrac{2}{60}F + \tfrac{9}{60}H + \tfrac{2}{60}M \le 100 \quad \text{(Test device 3 hours, all vars } \ge 0\text{)}$$

Adjustable Cells (Optimal Profit = \$195,504.83)

Cell	**Name**	**Final Value**	**Reduced Cost**	**Obj. Coeff.**	**Allow. Inc.**	**Allow. Dec.**
`$B$5`	Internal modems (I)	496.55	0.00	161.35	55.53	69.41
`$C$5`	External modems (E)	1,241.38	0.00	92.95	310.43	23.80
`$D$5`	Circuit boards (C)	0.00	-138.64	135.50	138.64	1E+30
`$E$5`	Floppy drives (F)	0.00	-57.44	82.50	57.44	1E+30
`$F$5`	Hard drives (H)	0.00	-221.73	249.80	221.73	1E+30
`$G$5`	Memory boards (M)	0.00	-269.86	191.75	269.86	1E+30

Constraints

Cell	**Name**	**Final Value**	**Shadow Price**	**RHS**	**Allow. Inc.**	**Allow. Dec.**
`$H$11`	Test device 1	120.00	1,284.52	120.00	47.83	48.00
`$H$12`	Test device 2	120.00	344.69	120.00	80.00	85.71
`$H$13`	Test device 3	62.07	0.00	100.00	1E+30	37.93

2. Answers to Sensitivity Questions

(a) Reduced Costs: Four products (C, F, H, and M) are not produced (`Final Value` $= 0$). Their reduced costs indicate the minimum amounts by which their unit profit contributions must increase before they would enter the optimal mix: **Circuit boards:** +\$138.64 (from \$135.50 to \$274.14); **Floppy drives:** +\$57.44 (from \$82.50 to \$139.94); **Hard drives:** +\$221.73 (from \$249.80 to \$471.53); **Memory boards:** +\$269.86 (from \$191.75 to \$461.61).

(b) Giving Up 35 Hours on Test Device 3: The current plan uses only 62.07 hours of the 100 available hours on device 3 (37.93 hours unused slack). Giving up 35 hours is within the allowable decrease of 37.93 hours (Shadow Price = \$0), so **Quitmeyer's optimal solution and profit will not be affected**.

(c) 20 Additional Hours on Test Device 1 at \$25/hr: Each additional hour on device 1 (up to 47.83 hours) increases profit by \$1,284.52. Because this shadow price already reflects the regular \$15/hr cost of device 1, paying \$25/hr incurs a premium of \$10/hr. The 20 hours increase profit by ($1,284.52 $\times$ 20) $-$ ($10 $\times$ 20) = **\$25,490.40**, yielding a new profit of **\$220,995.23**. **The deal is worthwhile.**

(d) Trading 20 Hours on Device 1 for 40 Hours on Device 2: Checking the **100% rule**: $(20/48) + (40/80) = 0.4167 + 0.50 = 0.9167 < 1$, so shadow prices are valid. The new profit is \$195,504.83 $-$ 20(\$1,284.52) + 40(\$344.69) = **\$183,602.03** (a loss of \$11,902.80). **The deal is not worthwhile.**

Problem 4-18: Battery Park Stable

Consider the Battery Park Stables animal feed problem first presented in Chapter 3 as Problem 3-34 on page 133. Use Solver to create the Sensitivity Report for this LP problem. Then answer the following questions using this report. Each question is independent of the others.

(a) If the price of grain decreases by \$0.01 per pound, will the optimal solution change?
(b) Which constraints are binding? Interpret the shadow price for the binding constraints.
(c) What would happen to the total cost if the price of mineral decreased by 20% from its current value?
(d) For what price range of oats is the current solution optimal?

Reference Data from Problem 3-34 (Page 133): Battery Park Stable mixes three feeds—an oat product (O), enriched grain (G), and mineral product (M)—to meet daily minimum requirements for five nutrients (A–E, in units/lb) at minimum cost, subject to a maximum daily feed limit (6 lbs/day in `P4-18.xls`; 5 lbs/day in later printings—non-binding in both cases since 4.67 lbs are used):

Nutrient / Cost	**Oat (O)**	**Grain (G)**	**Mineral (M)**	**Min. Needed (units/day)**
Ingredient A	2.0	3.0	1.0	6
Ingredient B	0.5	1.0	0.5	2
Ingredient C	3.0	5.0	6.0	9
Ingredient D	1.0	1.5	2.0	8
Ingredient E	0.5	0.5	1.5	5
Cost per lb	**\$0.09**	**\$0.14**	**\$0.17**	Max feed $\leq$ 6 lbs/day

Solution to Problem 4-18

1. LP Formulation and Excel Sensitivity Report (see `P4-18.xls`)

$$\text{Minimize Daily Feed Cost} = \$0.09O + \$0.14G + \$0.17M$$

subject to:

$$2.0O + 3.0G + 1.0M \geq 6 \quad \text{(Ingredient A)}$$
$$0.5O + 1.0G + 0.5M \geq 2 \quad \text{(Ingredient B)}$$
$$3.0O + 5.0G + 6.0M \geq 9 \quad \text{(Ingredient C)}$$
$$1.0O + 1.5G + 2.0M \geq 8 \quad \text{(Ingredient D)}$$
$$0.5O + 0.5G + 1.5M \geq 5 \quad \text{(Ingredient E)}$$
$$1.0O + 1.0G + 1.0M \leq 6 \quad \text{(Max feed, } O, G, M \geq 0\text{)}$$

Adjustable Cells (Optimal Cost = \$0.6867)

Cell	**Name**	**Final Value**	**Reduced Cost**	**Obj. Coeff.**	**Allow. Inc.**	**Allow. Dec.**
`$B$5`	Pounds Oat (O)	1.3333	0.0000	0.0900	0.0033	0.0050
`$C$5`	Pounds Grain (G)	0.0000	0.0050	0.1400	1E+30	0.0050
`$D$5`	Pounds Mineral (M)	3.3333	0.0000	0.1700	0.0100	0.1250

Constraints

Cell	**Name**	**Final Value**	**Shadow Price**	**RHS**	**Allow. Inc.**	**Allow. Dec.**
`$E$13`	Max feed	4.6667	0.0000	6.0	1E+30	1.3333
`$E$8`	Ingredient A	6.0000	0.0033	6.0	4.0000	2.0000
`$E$9`	Ingredient B	2.3333	0.0000	2.0	0.3333	1E+30
`$E$10`	Ingredient C	24.0000	0.0000	9.0	15.0000	1E+30
`$E$11`	Ingredient D	8.0000	0.0833	8.0	4.0000	0.8000
`$E$12`	Ingredient E	5.6667	0.0000	5.0	0.6667	1E+30

2. Answers to Sensitivity Questions

(a) Price of Grain Decreases by \$0.01 per Pound: The decrease of \$0.01/lb is beyond the allowable decrease limit of \$0.005/lb for Grain. Therefore, **the optimal solution will change**.

(b) Binding Constraints and Shadow Prices: The constraints for the minimum daily requirements of **Ingredient A** and **Ingredient D** are binding:

- **Ingredient A (Shadow Price** = \$0.0033**):** Each additional unit of Ingredient A required in the mix (up to 10 units) will increase the daily cost by **\$0.003** ($\frac{1}{3}$ cent).
- **Ingredient D (Shadow Price** = \$0.0833**):** Each additional unit of Ingredient D required in the mix (up to 12 units) will increase the daily cost by **\$0.083** (8.33 cents).

(c) Price of Mineral Decreases by 20%: A 20% decrease in the cost of mineral implies a new price of $0.80 \times \$0.17 = \$0.136$/lb, a decrease of \$0.034/lb. Because this is within the allowable decrease limit of \$0.125/lb, the current solution ($O = \frac{4}{3}$ lbs, $G = 0$ lbs, $M = \frac{10}{3}$ lbs) **remains optimal**, and the revised total cost is $\$0.09(\frac{4}{3}) + \$0.136(\frac{10}{3}) = \$0.5733 \approx \$\mathbf{0.57}$.

(d) Optimality Range for Price of Oats: With a current price of \$0.090/lb, allowable decrease of \$0.005, and allowable increase of \$0.0033, the price of oats can fluctuate between **\$0.085** and **\$0.093 per pound** (**\$0.0933/lb**) for the current solution to remain optimal.

Problem 4-19: Kathy Roniger's Campus Diet Problem

Consider Kathy Roniger's diet problem first presented in Chapter 3 as Problem 3-35 on page 133. Use Solver to create the Sensitivity Report for this LP problem. Then answer the following questions using this report. Each question is independent of the others.

(a) Interpret the shadow prices for the carbohydrates and iron constraints.
(b) What will happen to total cost if Kathy insists on using milk in her diet?
(c) What will be the maximum amount Kathy could pay for beans to make it a cost-effective item for inclusion in her diet?
(d) Is the solution to this problem a unique optimal solution? Justify your answer.

Solution to Problem 4-19

1. LP Formulation and Excel Sensitivity Report (see `P4-19.xls`)

From Problem 3-35 (page 133), the LP minimizes meal cost across Milk (M), Ground meat (G), Chicken (C), Fish (F), Beans (B), Spinach (S), and Potatoes (P):

$$\text{Minimize Cost} = 0.60M + 2.35G + 1.15C + 2.25F + 0.58B + 1.17S + 0.33P$$

subject to: Calories between 900 and 1,500; Iron ≥ 4 mg; Fat ≤ 50 gm; Protein ≥ 26 gm; Carbohydrates ≤ 50 gm; and all variables ≥ 0.

Adjustable Cells (Optimal Cost = \$1.7460 ≈ \$1.75)

Cell	Name	Final Value	Reduced Cost	Obj. Coeff.	Allow. Inc.	Allow. Dec.
B5	Pounds Milk (M)	0.0000	0.1484	0.60	1E+30	0.1484
C5	Pounds Ground meat (G)	0.4991	0.0000	2.35	0.2273	11.0640
D5	Pounds Chicken (C)	0.1728	0.0000	1.15	0.9729	0.0716
E5	Pounds Fish (F)	0.0000	1.2323	2.25	1E+30	1.2323
F5	Pounds Beans (B)	0.0000	0.2610	0.58	1E+30	0.2610
G5	Pounds Spinach (S)	0.1050	0.0000	1.17	0.2272	0.9800
H5	Pounds Potatoes (P)	0.7620	0.0000	0.33	0.4101	0.7534

Constraints

Cell	Name	Final Value	Shadow Price	RHS	Allow. Inc.	Allow. Dec.
I11	Max calories	900.00	0.0000	1,500	1E+30	600.00
I12	Max fat	50.00	-0.0033	50	3.74	7.21
I13	Max carbohydrates	50.00	-0.0070	50	11.07	44.19
I8	Min calories	900.00	0.0022	900	90.39	47.38
I9	Min iron	4.00	0.0745	4	33.24	1.38
I10	Min protein	60.78	0.0000	26	34.78	1E+30

2. Answers to Sensitivity Questions

(a) Shadow Prices for Carbohydrates and Iron:

- **Max carbohydrates (Shadow Price = −\$0.0070):** Each additional gram of carbohydrates allowed in the meal (up to $50 + 11.07 = 61.07$ gm) will **reduce** the meal cost by **\$0.007** (0.7 cents).
- **Min iron (Shadow Price = +\$0.0745):** Each additional milligram of iron required in the diet (up to $4 + 33.24 = 37.24$ mg) will **increase** the meal cost by **\$0.074** (\$0.0745, or 7.45 cents).

(b) Forcing Milk into the Diet: Milk has a reduced cost of \$0.1484. Therefore, each pound of milk used in the diet will cause the meal cost to **increase by \$0.148** (14.84 cents).

(c) Maximum Cost-Effective Price for Beans: Beans currently cost \$0.58 per pound. Looking at the reduced cost (and allowable decrease), the price of beans would have to decrease by at least \$0.261—to a maximum price of $\$0.580 - \$0.261 = \$\mathbf{0.319}$ **per pound**—before Kathy could consider including it in the meal.

(d) Uniqueness of the Optimal Solution: Yes, the current solution is a unique optimal solution. None of the allowable increase and allowable decrease values for the objective function coefficients is zero. Furthermore, all items that are currently not in the meal (milk, fish, and beans) have non-zero (strictly positive) reduced costs.

Problem 4-21: Mt. Sinai Hospital

Consider the Mt. Sinai Hospital example first presented in Chapter 3 as Problem 3-45 on page 136. Use Solver to create the Sensitivity Report for this LP problem. Then answer the following questions using this report. Each question is independent of the others.

(a) What is the maximum revenue per year, how many medical patients per year are there, and how many surgical patients per year are there? How many medical beds and how many surgical beds of the 90-bed addition should be added?
(b) Are there any empty beds with this optimal solution? If so, how many empty beds are there? Discuss the effect of acquiring more beds, if needed.
(c) Are the laboratories being used to their capacity? Is it possible to perform more lab tests per year? If so, how many more? Discuss the effect of acquiring more lab space, if needed.
(d) Is the x-ray facility being used to its maximum? Is it possible to do more x-rays per year? If so, how many more? Discuss the effect of acquiring more x-ray facilities, if needed.
(e) Is the operating room being used to capacity? Is it possible to do more operations per year? If so, how many more? Discuss the effect of acquiring more operating room, if needed.

Reference Data from Problem 3-45 (Page 136): Mt. Sinai Hospital is adding 90 beds ($90 \times 365 = 32{,}850$ patient-days/year). A medical patient (M) stays an average of 8 days, generates \$2,280 in revenue, requires 3.1 lab tests, 1 x-ray, and 0 surgeries. A surgical patient (S) stays an average of 5 days, generates \$1,515 in revenue, requires 2.6 lab tests, 2 x-rays, and 1 surgery. Annual capacities are 15,000 lab tests, 7,000 x-rays, and 2,800 surgeries.

Solution to Problem 4-21

1. LP Formulation and Excel Sensitivity Report (see `P4-21.xls`)

$$\text{Maximize Annual Revenue} = \$2{,}280M + \$1{,}515S$$

subject to:

$$\begin{aligned} 8M + 5S &\leq 32{,}850 && \text{(Patient-days: 90 beds} \times \text{365 days)} \\ 3.1M + 2.6S &\leq 15{,}000 && \text{(Lab tests capacity)} \\ 1M + 2S &\leq 7{,}000 && \text{(X-rays capacity)} \\ 1S &\leq 2{,}800 && \text{(Surgeries capacity, } M, S \geq 0\text{)} \end{aligned}$$

Adjustable Cells (Optimal Revenue = \$9,551,659.09)

Cell	Name	Final Value	Reduced Cost	Obj. Coeff.	Allow. Inc.	Allow. Dec.
B5	Medical patients (M)	2,790.91	0.00	2,280.00	144.00	1,522.50
C5	Surgical patients (S)	2,104.55	0.00	1,515.00	3,045.00	90.00

Constraints

Cell	Name	Final Value	Shadow Price	RHS	Allow. Inc.	Allow. Dec.
D8	Patient-days	32,850.00	276.82	32,850	2,677.78	7,650.00
D9	Lab tests	14,123.64	0.00	15,000	1E+30	876.36
D10	X-rays	7,000.00	65.45	7,000	956.25	2,893.75
D11	Oper/Surgeries	2,104.55	0.00	2,800	1E+30	695.45

2. Answers to Sensitivity Questions

(a) Optimal Patients, Bed Allocation, and Revenue:

- **Maximum Annual Revenue: \$9,551,659.09**

- **Medical Patients (M):** 2,790.91 $\approx$ **2,791 patients/year**, translating to $\frac{8\times 2{,}790.91}{365} = 61.17 \approx$ **61 medical beds**.
- **Surgical Patients (S):** 2,104.55 $\approx$ **2,105 patients/year**, translating to $\frac{5\times 2{,}104.55}{365} = 28.83 \approx$ **29 surgical beds**.

(b) Bed Utilization and Value of Additional Beds: There are **no empty beds** with this optimal solution (all 32,850 patient-days are used). Each additional patient-day (over the current 32,850) will permit Mt. Sinai to increase revenue by $\$$**276.82**. That is, by acquiring another bed (365 patient-days), annual revenue can be increased by $365 \times \$276.818 = \$$**101,039**.

(c) Laboratory Capacity: Labs are not used to capacity; they have an unused capacity of **876.364 lab tests/year**. Acquiring more lab space is therefore not worthwhile (Shadow Price = \$0).

(d) X-Ray Facility Capacity: X-ray capacity is being utilized to its **fullest extent** (7,000 of 7,000 x-rays). Each additional x-ray that can be handled (up to 956.25 x-rays) will increase annual revenue by $\$$**65.45**.

(e) Operating Room Capacity: The operating room is not used to capacity; it has an unused capacity of **695.45 operations/year**. Acquiring more operating room capacity is therefore not worthwhile (Shadow Price = \$0).

Problem 4-22: Good-to-Go Suitcase Company

The Good-to-Go Suitcase Company makes three kinds of suitcases: (1) Standard, (2) Deluxe, and (3) Luxury styles. Each suitcase goes through four production stages: (1) cutting and coloring, (2) assembly, (3) finishing, and (4) quality and packaging. The total number of hours available in each of these departments is 630, 600, 708, and 135, respectively.

Each Standard suitcase requires 0.7 hours of cutting and coloring, 0.5 hours of assembly, 1 hour of finishing, and 0.1 hours of quality and packaging. The corresponding numbers for each Deluxe suitcase are 1 hour, 0.83 hours, 0.67 hours, and 0.25 hours, respectively. Likewise, the corresponding numbers for each Luxury suitcase are 1 hour, 0.67 hours, 0.9 hours, and 0.4 hours, respectively.

The sales revenue for each type of suitcase is as follows: Standard \$36.05, Deluxe \$39.50, and Luxury \$43.30. The material costs are Standard \$6.25, Deluxe \$7.50, and Luxury \$8.50. The hourly cost of labor for each department is cutting and coloring \$10, assembly \$6, finishing \$9, and quality and packaging \$8.

The Excel layout and LP Sensitivity Report of Good-to-Go's problem are shown in Screenshots 4-10A and 4-10B, respectively. Answer the following questions, each of which is independent of the others.

Screenshot 4-10A: Excel Layout for Problem 4-22 (Good-to-Go Suitcase Company)

Row / Constraint	Standard	Deluxe	Luxury	Total / LHS	Sign	RHS	Cost/hr
Solution value	540.00	252.00	0.00				
Selling price per unit	\$36.05	\$39.50	\$43.30	\$29,421.00			
Material cost per unit	\$6.25	\$7.50	\$8.50	\$5,265.00			
Labor cost per unit	\$19.80	\$23.00	\$25.30	\$16,488.00			
Profit per unit	**\$10.00**	**\$9.00**	**\$9.50**	**\$7,668.00**			
Cutting & Coloring	0.70	1.00	1.00	630.00	$\leq$	630	\$10
Assembly	0.50	0.83	0.67	480.00	$\leq$	600	\$6
Finishing	1.00	0.67	0.90	708.00	$\leq$	708	\$9
Quality & Packaging	0.10	0.25	0.40	117.00	$\leq$	135	\$8

Screenshot 4-10B: Solver Sensitivity Report for Problem 4-22

Adjustable Cells						
Cell	**Name**	**Final Value**	**Reduced Cost**	**Obj. Coeff.**	**Allow. Inc.**	**Allow. Dec.**
`$B$4`	Solution value Standard	540.00	0.00	10.00	3.50	2.56
`$C$4`	Solution value Deluxe	252.00	0.00	9.00	5.29	1.61
`$D$4`	Solution value Luxury	0.00	-1.12	9.50	1.12	1E+30
Constraints						
Cell	**Name**	**Final Value**	**Shadow Price**	**RHS**	**Allow. Inc.**	**Allow. Dec.**
`$E$10`	Cutting & Coloring	630.00	4.38	630	52.36	134.40
`$E$11`	Assembly	480.00	0.00	600	1E+30	120.00
`$E$12`	Finishing	708.00	6.94	708	192.00	126.00
`$E$13`	Quality & Packaging	117.00	0.00	135	1E+30	18.00

(a) What is the optimal production plan? Which of the resources are scarce?

(b) Suppose Good-to-Go is considering including a polishing process, the cost of which would be added directly to the price. Each Standard suitcase would require 10 minutes of time in this treatment, each Deluxe suitcase would need 15 minutes, and each Luxury suitcase would need 20 minutes. Would the current production plan change as a result of this additional process if 170 hours of polishing time were available? Explain your answer.

(c) Now consider the addition of a waterproofing process in which each Standard suitcase would use 1 hour of time in the process, each Deluxe suitcase would need 1.5 hours, and each Luxury suitcase

would require 1.75 hours. Would this change the production plan if 900 hours were available? Why or why not?

Solution to Problem 4-22

(a) Optimal Production Plan and Scarce Resources: The optimal production plan is to produce **540 Standard suitcases** and **252 Deluxe suitcases** (and **0 Luxury suitcases**), for a total profit of **$7,668.00**. The scarce (binding) resources are **Cutting & Coloring** (630 hours) and **Finishing** (708 hours).

(b) Addition of Polishing Process (170 Hours Available): The polishing constraint (in hours) is $\frac{1}{6}(\text{Standard}) + \frac{1}{4}(\text{Deluxe}) + \frac{1}{3}(\text{Luxury}) \leq 170$. Substituting the current optimal production plan yields $\frac{1}{6}(540) + \frac{1}{4}(252) + \frac{1}{3}(0) = 90 + 63 + 0 = 153$ hours ≤ 170 hours. Therefore, there is sufficient capacity in the proposed polishing operation to sustain the optimal production plan, and **the production plan will not change**.

(c) Addition of Waterproofing Process (900 Hours Available): The waterproofing constraint (in hours) is $1(\text{Standard}) + 1.5(\text{Deluxe}) + 1.75(\text{Luxury}) \leq 900$. Substituting the optimal production plan yields $1(540) + 1.5(252) + 1.75(0) = 540 + 378 + 0 = 918$ hours > 900 hours. Therefore, the proposed waterproofing process does not have sufficient capacity to support the current plan. **Yes, the production plan will change**; to determine the exact impact, we must return to the original LP formulation and add this constraint.

Problem 4-24: Strollers-to-Go Company

The Strollers-to-Go Company makes lightweight umbrella-type strollers for three different groups of children: TiniTote (for newborns), ToddleTote (for toddlers up to 30 pounds), and TubbyTote (heavy-duty model up to 60 pounds). Marketing has forecast maximum demand of 180 TiniTotes (selling for \$63.75), 70 TubbyTotes (\$82.50), and 160 ToddleTotes (\$66.00). Policy requires producing at least 50% of forecast demand for each model and keeping ToddleTotes to at most 40% of total stroller production. Material costs are \$4.00, \$6.00, and \$5.50 per unit, respectively. Fabrication has 620 hours available at \$8.25/hr; sewing has 500 hours at \$8.50/hr; assembly has 480 hours at \$8.75/hr. TiniTotes require 3, 2, and 1 hours in fabrication, sewing, and assembly; TubbyTotes require 4, 1, and 3 hours; ToddleTotes require 2 hours in each station.

The Excel layout and LP Sensitivity Report are shown in Screenshots 4-11A and 4-11B. Answer the following independent questions:

Screenshot 4-11A: Excel Layout for Problem 4-24 (Strollers-to-Go Company)

Row / Constraint	TiniTote	TubbyTote	ToddleTote	Total / LHS	Sign	RHS	Cost/hr
Solution value	100.00	35.00	90.00				
Selling price per unit	\$63.75	\$82.50	\$66.00	\$15,202.50			
Material cost per unit	\$4.00	\$6.00	\$5.50	\$1,105.00			
Labor cost per unit	\$50.50	\$67.75	\$51.00	\$12,011.25			
Profit per unit	**\$9.25**	**\$8.75**	**\$9.50**	**\$2,086.25**			
Fabrication	3.0	4.0	2.0	620.00	$\leq$	620	\$8.25
Sewing	2.0	1.0	2.0	415.00	$\leq$	500	\$8.50
Assembly	1.0	3.0	2.0	385.00	$\leq$	480	\$8.75
TiniTote demand	1.0	0.0	0.0	100.00	$\leq$	180	
TubbyTote demand	0.0	1.0	0.0	35.00	$\leq$	70	
ToddleTote demand	0.0	0.0	1.0	90.00	$\leq$	160	
ToddleTote max prod ratio	-0.4	-0.4	0.6	0.00	$\leq$	0	
TiniTote min prod	1.0	0.0	0.0	100.00	$\geq$	90	
TubbyTote min prod	0.0	1.0	0.0	35.00	$\geq$	35	
ToddleTote min prod	0.0	0.0	1.0	90.00	$\geq$	80	

Screenshot 4-11B: Solver Sensitivity Report for Problem 4-24

Adjustable Cells

Cell	Name	Final Value	Reduced Cost	Obj. Coeff.	Allow. Inc.	Allow. Dec.
\$B\$4	Solution value TiniTote	100.00	0.00	9.25	5.00	3.33
\$C\$4	Solution value TubbyTote	35.00	0.00	8.75	4.10	1E+30
\$D\$4	Solution value ToddleTote	90.00	0.00	9.50	1E+30	3.33

Constraints

Cell	Name	Final Value	Shadow Price	RHS	Allow. Inc.	Allow. Dec.
\$E\$10	Fabrication	620.00	3.60	620	110.50	43.33
\$E\$11	Sewing	415.00	0.00	500	1E+30	85.00
\$E\$12	Assembly	385.00	0.00	480	1E+30	95.00
\$E\$13	TiniTote demand	100.00	0.00	180	1E+30	80.00
\$E\$14	TubbyTote demand	35.00	0.00	70	1E+30	35.00
\$E\$15	ToddleTote demand	90.00	0.00	160	1E+30	70.00
\$E\$16	ToddleTote max prod ratio	0.00	3.95	0	13.00	8.67
\$E\$17	TiniTote min prod	100.00	0.00	90	10.00	1E+30
\$E\$18	TubbyTote min prod	35.00	-4.10	35	8.13	35.00
\$E\$19	ToddleTote min prod	90.00	0.00	80	10.00	1E+30

(a) How many strollers of each type should Strollers-to-Go make? What is the profit? Which constraints are binding?

(b) How much labor time is being used in the fabrication, sewing, and assembly areas?
(c) How much would Strollers-to-Go be willing to pay for an additional hour of fabrication time? For an additional hour of sewing time?
(d) Is Strollers-to-Go producing any product at its maximum sales level? Is it producing any product at its minimum level?

Solution to Problem 4-24

(a) Optimal Production Plan, Profit, and Binding Constraints: The optimal plan is to make **100 TiniTote**, **35 TubbyTote**, and **90 ToddleTote** strollers, yielding a profit of **$2,086.25**. The **binding constraints** are **fabrication time** (620 hrs), the **minimum production level for the TubbyTote model** (35 units), and the **ratio of the ToddleTote model to total production** ($90/225 = 40\%$).

(b) Labor Time Used: All **620 hours** are used in Fabrication; **415 of 500 hours** are used in Sewing; and **385 of 480 hours** are used in Assembly.

(c) Willingness to Pay for Fabrication and Sewing Time: Each additional hour of fabrication time (up to 110.50 hours) increases profit by **$3.60**, so the firm would pay a **premium of up to $3.60/hr** over the regular $8.25/hr rate (up to $11.85/hr total). Since sewing is non-binding (85 hrs slack, Shadow Price = $0), the firm would pay **$0** for additional sewing time.

(d) Maximum and Minimum Production Levels: **None** of the products is produced at its maximum demand level. However, the **TubbyTote** model is produced at its minimum required level (35 units).